Week 8 – Workshop
Geothermics: Steady-State Heat Flow

Adelaide University

Geophysics 3A: Potential Fields & Geothermics

July 31, 2026

Preparation

Pre-reading: Course Notes, Ch. 10
Lecture videos:
Notes:

Purpose

This workshop uses conservation-of-energy reasoning to build intuition for steady-state geotherms: how thermal conductivity contrasts refract isotherms and redistribute gradients, and how crustal heat production controls geotherm curvature and deep temperatures.

Learning Outcomes

By the end of this workshop, students should be able to:

- Use conservation of energy to reason about steady-state geotherms.
- Explain how thermal conductivity contrasts redistribute isotherms.
- Interpret geotherm curvature in terms of crustal heat production.
- Relate conductivity and heat production variations to geothermal potential.

Session Flow (?? min)

0–15 min: Warm-up 8.A: Thermal Refraction

Compare two crustal columns with identical basal heat flow:

- Case A: crystalline crust exposed at the surface.
- Case B: crystalline crust overlain by a low-conductivity sedimentary layer.

In this activity, you'll look at how temperature, thermal gradient and heat flow are affected by the presence of a sedimentary basin.

15–30 min: Mini-lecture — Steady-State Regime

- Steady-state heat flow as an energy bookkeeping problem.
- Review Fourier's law and the meaning of heat flow.
- Emphasize the separation of roles:
  - conductivity controls *gradients*,
  - heat production controls *curvature*.

Geological Discussion

- Implications for geothermal gradients measured in boreholes.
- Relative geothermal potential of sedimentary basins vs exposed cratons.

30–50 min: Activity 8.B: Structures and Steady-State Fields

In this problem, you will explore the temperature and heat flow patterns expected in the subsurface as a result of bodies with different geometries.

50–80 min: Activity 8.C: Layered and Anisotropic Conductivity

Anisotropy in conductivity manifests from structural arrangement of two or more materials. It can happen in layered rocks or layered sequences of rocks. You will explore the mathematical theory that gives rise to anisotropy and look at the anisotropy in several metamorphic hand samples.

80–85 min: Break

85–120 min: Activity 8.D: Heat Production from Seismic Velocity?

The dominant control on heat production is lithology. In this activity, you will look at whether the relationship is strong enough that seismic velocity can be used to predict heat production. The premise is based on research conducted in the late 1970's to 1990, but there are a lot more data available to us now than researchers did then. Let's see how their idea holds up under scrutiny.

120–155 min: Activity 8.E: Building a Geotherm

Once we've covered thermal conductivity and heat production and understand these important thermal properties better we will be ready to estimate geotherms. In this activity we will compute geotherms and see what happens when we change the thermal conductivity and heat production.

155–165 min: Wrap-up

- Summarize roles of k and A.
- Connect observations to upcoming steady-state problem set.

165–175 min: Preview: Transient Heat Flow

So far we have assumed that temperature does not change with time—a useful idealization for mature, stable terranes. They can even be used as reasonable approximations for snapshots of many active tectonic regions if they don't have transients that are too large. But many geologically interesting situations involve heat flow that evolves: cooling of the oceanic lithosphere, heating around a magmatic intrusion, or the thermal relaxation of a thrust belt. Next week we move to transient heat conduction, governed by the diffusion equation, and develop the concept of thermal diffusivity as the quantity that controls how quickly temperature anomalies propagate through rock.

Workshop Notes

Workshop Notes

Warm-up 8.A
Thermal Refraction

Purpose

To build intuition for how low-conductivity surface layers redistribute temperature gradients without changing the total heat flow.

Learning Outcomes

By the end of this activity, you should be able to:

- Apply Fourier's law to explain why the temperature gradient differs between regions with constant heat flow.
- Sketch temperature–depth profiles for a layered crust and identify where isotherms are compressed or stretched.
- Predict which crustal section reaches higher interface temperatures and explain why.
- Discuss how varying conductivity affects geothermal energy potential.

Geological Context

Consider two continental regions with identical basal heat flow (Figure **??**):

Region A: crystalline crust exposed at the surface.
Region B: crystalline crust overlain by a thick sedimentary basin.

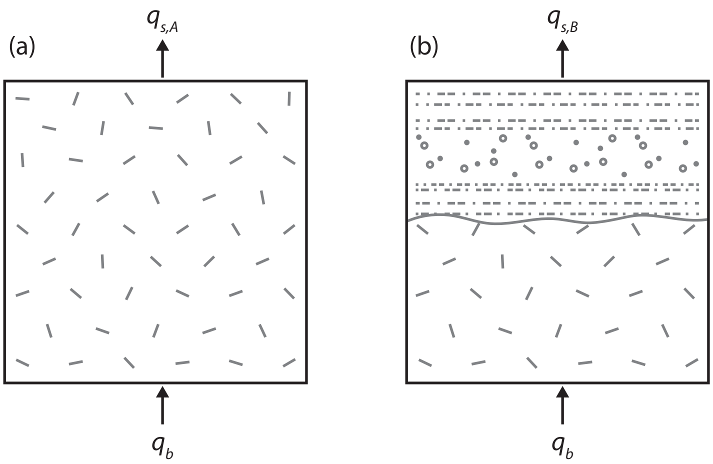


Figure 1: Two continental regions: (a) igneous crust, (b) sedimentary basin over igneous crust. Both regions have the same basal heat flow.

The sediments are dominated by shales and porous clastic rocks with lower thermal conductivity than basement rock. We will make the following assumptions:

- no internal heat production,
- steady-state heat flow, and
- heat flow is vertical and laterally uniform.

Tasks

Question 1. Reasoning

What is the relationship between the surface heat flow of regions A and B? What principle did you use to make this determination?

Using Fourier's law,

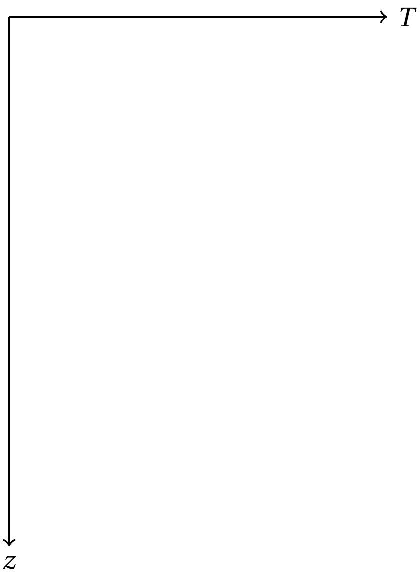
$$q = -k\frac{dT}{dz},$$

explain qualitatively how the temperature gradient differs between the sedimentary basin and the igneous basement?

Question 2. Sketch

Sketch temperature versus depth for both regions on the same axes. Clearly label:

- sediment thickness,
- basement,
- relative steepness of gradients.



Question 3. Explanation & Discussion

i Which region would exhibit a higher temperature at the sediment–basement interface?

Explain why.

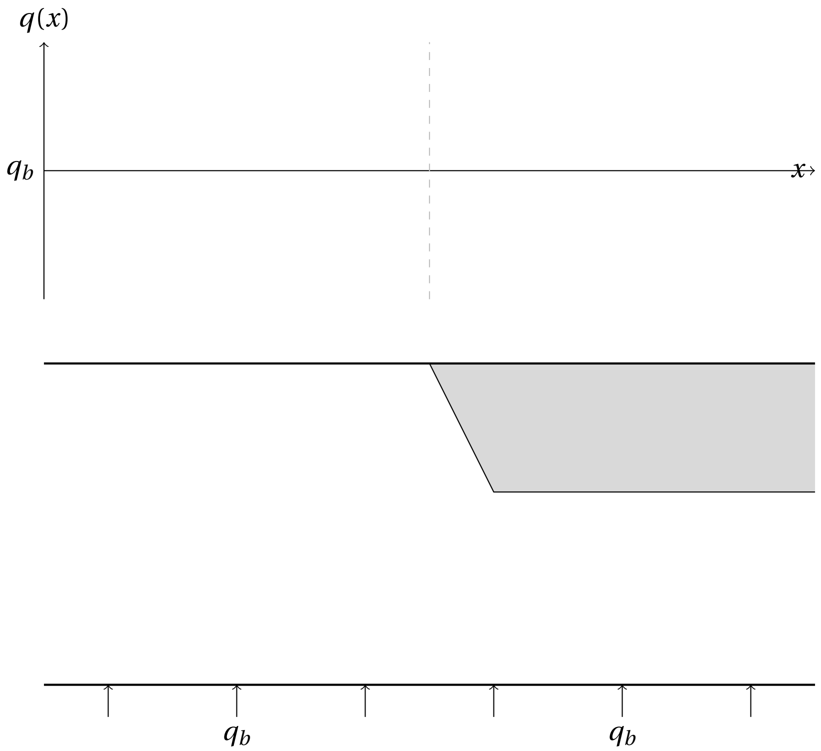
ii How does the presence of sediments affect measured geothermal gradients?

iii Which region would you expect to have higher geothermal potential? Why?

Question 4. Extension

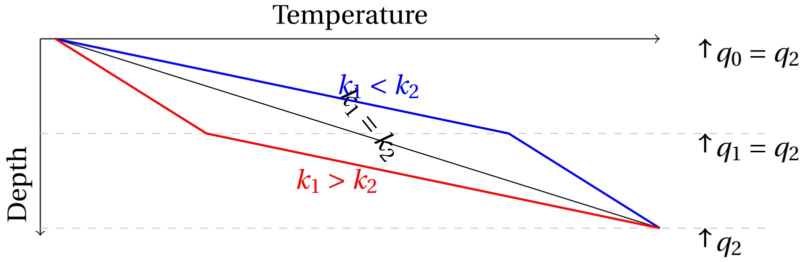
If these two regions were adjacent to each other, how would heat flow and temperature vary in 2D?

Start by drawing the heat flow lines, then the isotherms and finally the variation in surface heat flow on the axes provided.



Key Ideas

A conductivity contrast redistributes the gradient, not the flux. Whenever q must stay constant across a boundary but k changes, $dT/dz = -q/k$ forces the gradient to change instead. This effect is thermal refraction, and it applies at any conductivity contrast, not only a sediment-over-basement boundary; the same reasoning will reappear for layered media, tilted strata, and mixed lithologies.



A measured gradient reflects local conductivity, not the deeper structure. The same heat flow can correspond to very different gradients depending on what you're measuring through, so gradients from different settings are not directly comparable without calculation of heat flow. A gradient measured near the surface is only converted to a true heat flow once you know the conductivity at the depth it was measured.

Activity 8.B
Structures and Steady-State Fields

Purpose

Gravity and magnetics activities in this course have had you sketch fields around simple structures—a sphere, a dike, a topographic high—and reason about the pattern from first principles before any computation. This activity does the same for heat conduction: given a structure of different conductivity than its surroundings, sketch the isotherms (temperature field) and heat flux lines (analogous to field lines) around it, in steady state, with no internal heat production anywhere.

Learning Outcomes

By the end of this activity, you should be able to:

- Sketch how isotherms and flux lines distort around a body of different thermal conductivity than its host.
- Explain why flux lines bend toward the higher-conductivity path, and connect this to the same refraction principle already used for electrical current and (in Week 6) magnetic field lines.
- Distinguish a field distorted by a *material property* contrast (conductivity) from one distorted by a *boundary geometry* effect (topography), using the same steady-state machinery for both.

Background

In steady state with no internal heat production, two conditions hold at any interface between different conductivities:

- Temperature is continuous across the boundary (no physical jump).
- The heat flux *normal* to the boundary is continuous (nothing is accumulating there).

Just as electrical current refracts at a resistivity contrast, and magnetic field (B) lines refract at a susceptibility contrast, heat flux lines refract at a conductivity contrast—bending toward whichever side lets more flux pass for the same driving gradient. None of the three structures below has any internal heat production; the only thing distorting each field is either a conductivity contrast or, in the third case, the shape of the surface itself.

Tasks

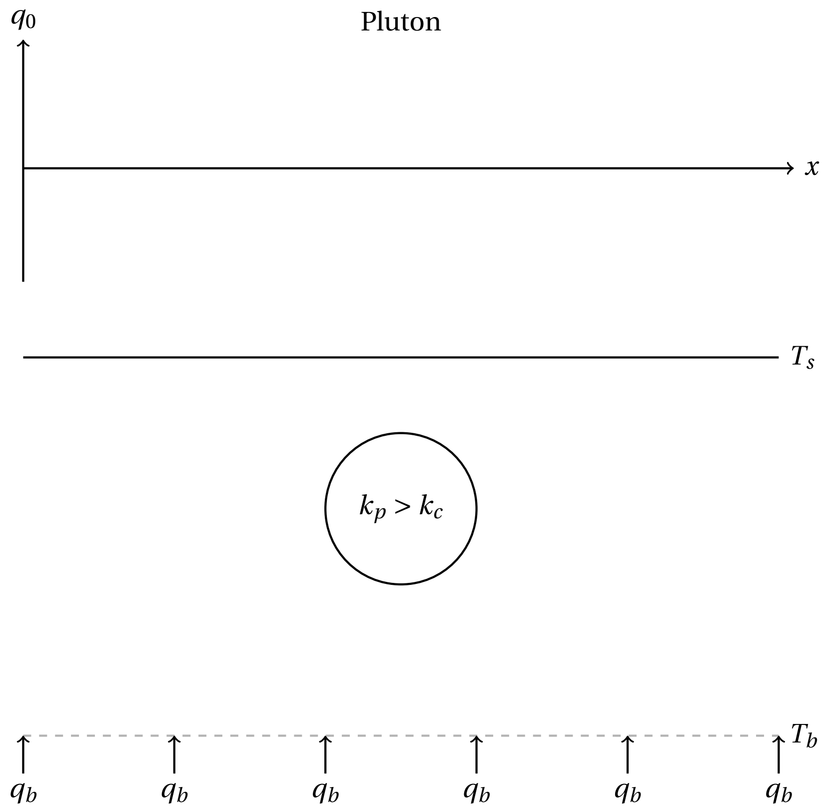
Question 1. Temperature vs. Heat Flux

What is the mathematical relationship between heat flow and temperature?

What is the geometrical relationship between isotherms (lines of constant temperature) and heat flow lines?

Question 2. A Pluton

A circular pluton of higher conductivity, k_p, than its host sits in otherwise uniform crust, k_c, far from any other structure. Regional heat flow q points toward the surface, undisturbed far from the pluton (dashed lines).



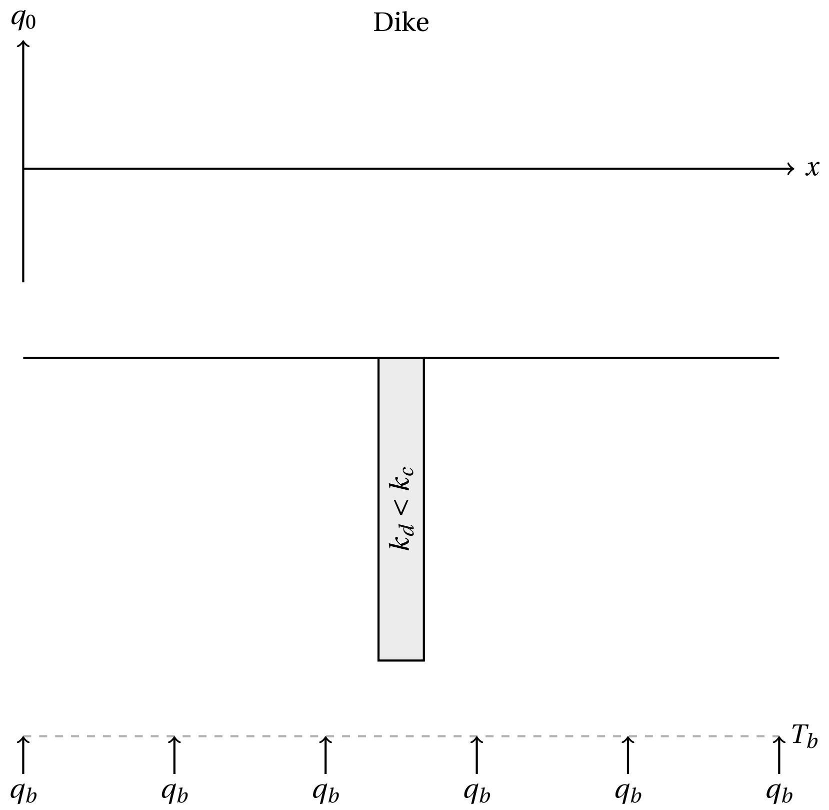
(a) Sketch flux lines through the whole panel, including through and around the pluton. Do they concentrate into the pluton, or avoid it?

(b) Sketch isotherms consistent with your flux lines. Where are they compressed (steep gradient), and where are they spread out (gentle gradient)?

(c) If k_p were *less* than k_c instead, without redrawing, describe how your answer to (a) would change.

Question 3. A Dike

A near-vertical dike of lower conductivity, k_d than its host cuts across the same regional field, k_c.

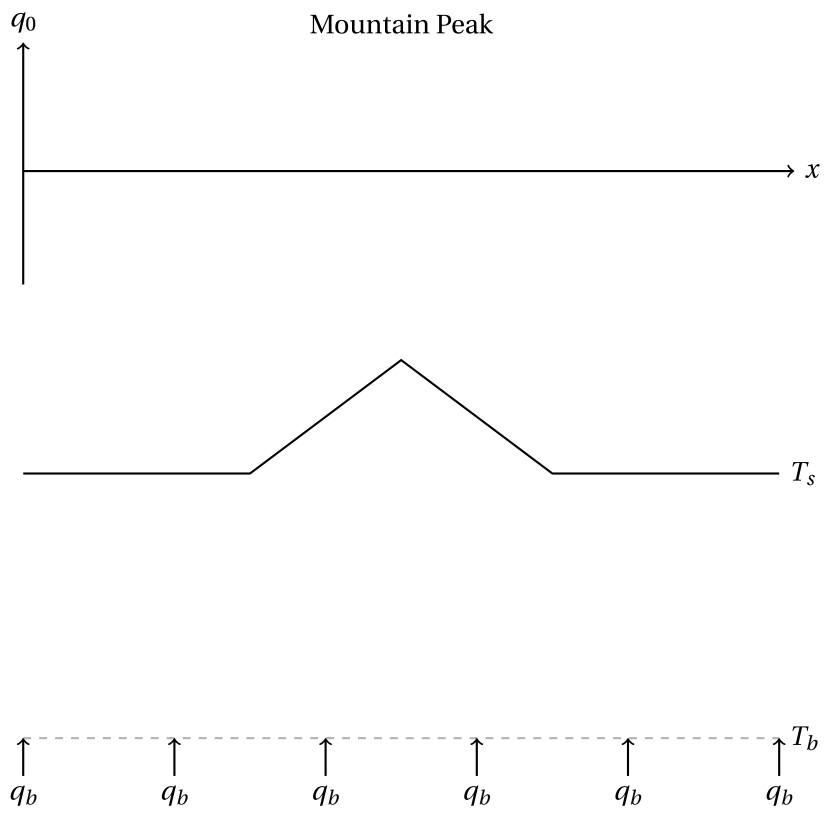


(a) Sketch flux lines through the panel. Does flux concentrate into the dike or avoid it? How does this compare to the sphere above?

(b) Sketch isotherms consistent with your flux lines. Would a borehole drilled straight down just outside the dike measure a different near-surface gradient than one drilled far away?

Question 4. A Triangular Mountain

Uniform crust, uniform conductivity throughout, but Earth's surface itself is not flat. Deep, undisturbed isotherms are shown; the surface follows the topographic profile. Sketch isotherms from the deep, undisturbed region up through the topography, including beneath the peak itself.



(a) Do the isotherms stay flat all the way to the surface, or bend to follow the topographic profile?

(b) This distortion has nothing to do with a conductivity contrast since conductivity is uniform everywhere in this panel. What, physically, is causing the isotherms to bend?

(c) At what distance from the peak (roughly, in mountain-heights) would you expect this distortion to have mostly died away? Compare your reasoning to how gravity and magnetic anomalies decay with distance from a source.

Key Ideas

Heat-flux lines refract at a thermal conductivity contrast because the temperature and heat flux must satisfy continuity conditions across the interface. Temperature is continuous across the boundary, and the normal component of heat flux is conserved in the absence of internal heat sources or sinks. Since Fourier's law, $\mathbf{q} = -k\nabla T$, relates heat flux to the temperature gradient, a change in thermal conductivity requires the temperature gradient to change direction at the interface. As a result, heat-flux lines bend, or refract, across the boundary.

Flux lines refract at a conductivity contrast, always bending toward the path of least resistance. Whether flux concentrates into or avoids a structure depends on whether that structure is higher or lower conductivity than its host. A pluton (high k) draws flux in; a dike (low k, as drawn here) pushes flux around it. Reversing the conductivity contrast reverses the pattern.

Not every field distortion comes from a material contrast. The mountain's isotherms bend because the boundary condition itself (the surface, where $T = T_0$) is not flat. This phenomena is the same kind of geometric effect that produces terrain corrections in gravity and magnetics, distinct from (and addable to) any density, susceptibility, or conductivity contrast in the subsurface.

Activity 8.C
Layered and Anisotropic Conductivity

Purpose

To see how anisotropy of thermal conductivity arises from simple geometric models of layered materials and how this is expressed in natural samples.

Learning Outcomes

By the end of this activity, you should be able to:

- Derive the effective conductivity of layered media when heat flows parallel or perpendicular to layering.
- Use real directional conductivity data to estimate an anisotropy factor and propose a functional form for how conductivity varies with angle.

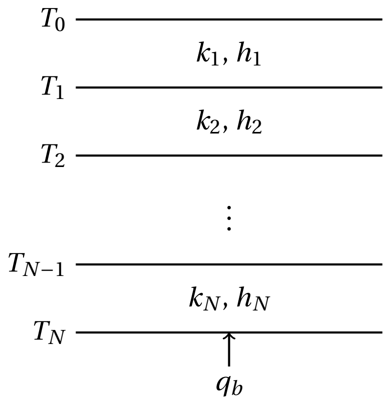
Geological Context

Layering can happen in many geological contexts and at all scales from sheets of mica to layers in a sedimentary basin. This activity builds the theoretical bounds on that directional dependence, then checks them against real measurements.

Tasks

Question 1. Perpendicular Case: Heat Flow Across the Layering

Consider N layers stacked with heat flowing perpendicular to the layering, crossing every layer in turn, for an arbitrary number of layers rather than just two. Layer i has thickness h_i and conductivity k_i; the interface temperatures are T_0 (top) through T_N (base), and $H = \sum_i h_i$.



(a) In steady state with no internal heat production, what must be true about q at every interface between layers, and why?

(b) Heat flows upward, from T_N at the base toward T_0 at the top, so $T_N > T_0$: temperature increases with depth. Write Fourier's law for the temperature drop $\Delta T_i = T_i - T_{i-1}$ across layer i alone (positive, since the deeper interface is hotter), in terms of q, k_i, and h_i.

(c) The total drop across the whole stack, in the direction heat is flowing, is $\Delta T = T_N - T_0$. How does ΔT relate to the individual ΔT_i?

(d) Combine (b) and (c) to write ΔT as a single sum over all N layers.

(e) Define an effective conductivity $k_\perp$ by requiring the whole stack obey Fourier's law as if it were one layer: $\Delta T = qH/k_\perp$. Set this equal to your result from (d) and solve for $1/k_\perp$.

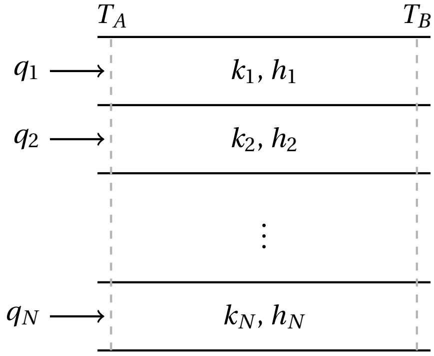
You should arrive at

$$\frac{1}{k_\perp} = \sum_{i=1}^{N} \frac{f_i}{k_i}, \qquad f_i \equiv \frac{h_i}{H}. \tag{1}$$

This function is the thickness-weighted *harmonic mean*, the same result as for two layers. The result falls out of the derivation for arbitrary N with no extra work, since the sum in (d) was already general.

Question 2. Parallel Case: Heat Flow Along the Layering

Now consider the *same* stack of layers, but with heat flowing along the layering instead, parallel to strike, not across it. Each layer independently carries its own share of the heat flow, q_i between two shared end temperatures T_A and T_B.



(a) What quantity is now shared by every layer? Not q this time, but something else. (Compare to part (a) of Question 1: there, q was forced to be equal across every layer. What plays that role here?)

(b) Heat flows from T_A toward T_B, so $T_A > T_B$. Write Fourier's law for the heat flow q_i carried by layer i, in terms of the shared gradient $g = (T_A - T_B)/\Delta x$ (positive, in the direction heat is flowing) and k_i.

(c) Each layer contributes $q_i h_i$ (its flux times its own thickness) to the total heat flow through the stack's cross-section. Write Q_{total} as a sum over all N layers.

(d) Define $k_{\parallel}$ by requiring the whole stack carry $Q_{\text{total}} = k_{\parallel}\, g\, H$. Solve for $k_{\parallel}$.

You should arrive at

$$k_{\parallel} = \sum_{i=1}^{N} f_i k_i, \tag{2}$$

the thickness-weighted *arithmetic mean* and notice this one required no manipulation of reciprocals at all; the sum was already in the right form from step (c).

Question 3. Numeric Check

A layered sequence has the properties below.

Layer	1	2	3	4
h_i (m)	12	4	20	8
k_i ($\mathrm{W\,m^{-1}K^{-1}}$)	1.2	7.0	1.5	5.0

(a) Compute $k_{\perp}$ (perpendicular, across-layering) for this sequence.

(b) Compute $k_\parallel$ (parallel, along-layering) for the same sequence.

(c) Which is larger? By roughly what factor?

(d) This ordering, $k_\parallel \geq k_\perp$, holds for *any* set of positive k_i — it's the arithmetic-mean/harmonic-mean inequality, a general mathematical fact, not a coincidence of these numbers. Physically, why should heat flowing *along* the layering always manage a higher effective conductivity than heat forced *across* it? (Hint: think about how much choice the heat has in each case about which layers carry it.)

Question 4. Real Data: Directional Conductivity in Foliated Rocks

Many metamorphic rocks develop a strong planar fabric—foliation—during deformation, aligning platy minerals (especially micas) into a preferred orientation.

The figure below shows measured thermal conductivity versus the angle between the measurement direction and foliation, for four samples: three metapelites and one gneiss.

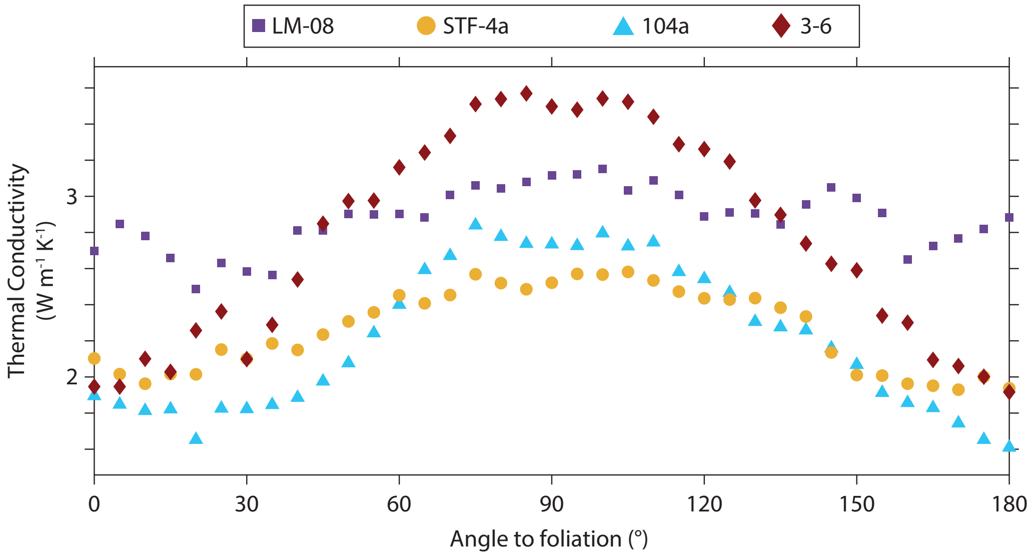


Figure 2: Thermal conductivity vs. angle to foliation for three metapelites and one gneiss.

(a) At 0°, is the measurement direction along the foliation plane or across it? What about 90°? State your reasoning – don't just guess the convention.

(b) Using your answer to (a) together with Questions 1 and 2, which angle should correspond to $k_{\parallel}$ and which to $k_{\perp}$?

(c) Does the real data fall between the bounds you'd expect, or does something surprise you? Where do the curves peak?

(d) Define the **anisotropy factor** as $A = k_{\max}/k_{\min}$ for a given sample. Estimate A for each of the four samples by reading approximate maximum and minimum conductivities off the figure.

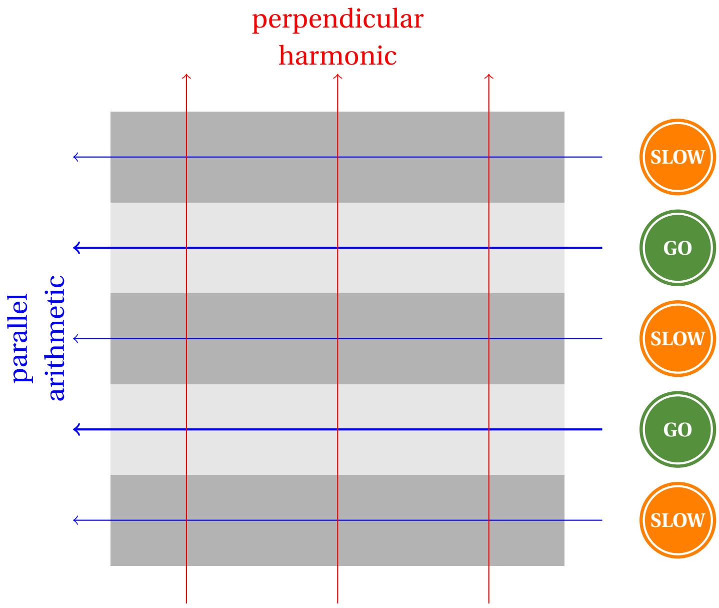
(e) Compare A across the three metapelites and the one gneiss. Is one rock type consistently more anisotropic than the other? Suggest a reason, connected to how each rock's fabric actually formed.

(f) Propose a simple mathematical function $k(\theta)$ that (i) equals your two derived bounds at the two extreme angles, and (ii) varies smoothly in between. You don't need to fit it precisely – just propose a reasonable functional form and justify why you chose it.

Key Ideas

The same physical setup gives different effective properties depending on flow direction. Heat flowing across layering shares the same flux and sums temperature drops (harmonic mean); heat flowing along layering shares the same gradient and sums fluxes (arithmetic mean). Recognizing *which quantity is common* to every layer is the whole derivation—the mean falls out mechanically once that's identified.

Parallel flow can always exploit the fastest pathway; perpendicular flow cannot avoid the slowest one. $k_{\parallel} \geq k_{\perp}$ always, because along-layering heat is free to concentrate disproportionately in the highest-conductivity layers, while across-layering heat has no choice but to cross every layer, including the most resistive one, at the same rate as all the others.



Real anisotropic media are not obligated to sit inside an idealized two-direction bound. The harmonic/arithmetic bracket comes from treating a rock as alternating layers of two conductivities. Real foliated rocks carry additional anisotropy from the crystal-scale conductivity of their constituent minerals, so the true directional dependence is a combination of layering geometry and mineral fabric, not layering geometry alone.

Activity 8.D
Heat Production from Seismic Velocity?

Purpose

Unlike many problem we have investigated thus far, heat production cannot be remotely sense $A(z)$ directly. This activity asks whether seismic velocity, which *can* be estimated at depth almost everywhere, can stand in as a proxy for heat production.

Learning Outcomes

By the end of this activity, you should be able to:

- Explain why heat production is harder to estimate at depth than density, susceptibility, or resistivity.
- Evaluate a real scientific critique of a proposed geophysical proxy relationship, and identify when a change of variable (not new data) resolves it.
- Use real velocity-depth data and an empirical velocity-heat production relationship to estimate a terrane's radiogenic crustal heat flow.
- Test that estimate against an independent constraint—conservation of energy—and reason about which of two disagreeing methods deserves more trust, and why.

Geological Context

Density, magnetic susceptibility, electrical resistivity, and seismic velocity can all, in principle, be estimated at depth from a surface survey – that is the entire premise of applied geophysics. Heat production cannot: it has no directly observable geophysical field of its own. Rybach and Čermák proposed a workaround decades ago – since seismic velocity depends on bulk mineralogy and heat production depends on trace radioactive element concentrations, and both broadly track a rock's felsic-to-mafic character, velocity might serve as an indirect proxy for heat production, remotely sensed the way heat production itself cannot be.

Fountain (1986) pushed back hard: plotted directly, A against V_p is extremely scattered – visually close to no relationship at all. This activity works through both Fountain's objection and a resolution to it, using two Australian terranes with very different histories: the **Macquarie Arc**, a Palaeozoic magmatic arc within the Lachlan Orogen, and the **Yilgarn Craton**, an Archean shield.

Tasks

Question 1. Linear or Log? Revisiting Fountain's Critique

Below is measured heat production plotted against P-wave velocity for both terranes, first in linear space, then in $\log_{10}$ space with the fitted relationship $\log_{10} A = m_0(V_p - 6.0) + m_1$ overlaid.

(a) Looking only at Figure **??**, would you say Fountain had a point?

(b) Now look at Figure **??**. Has the underlying data changed, or just how it's displayed?

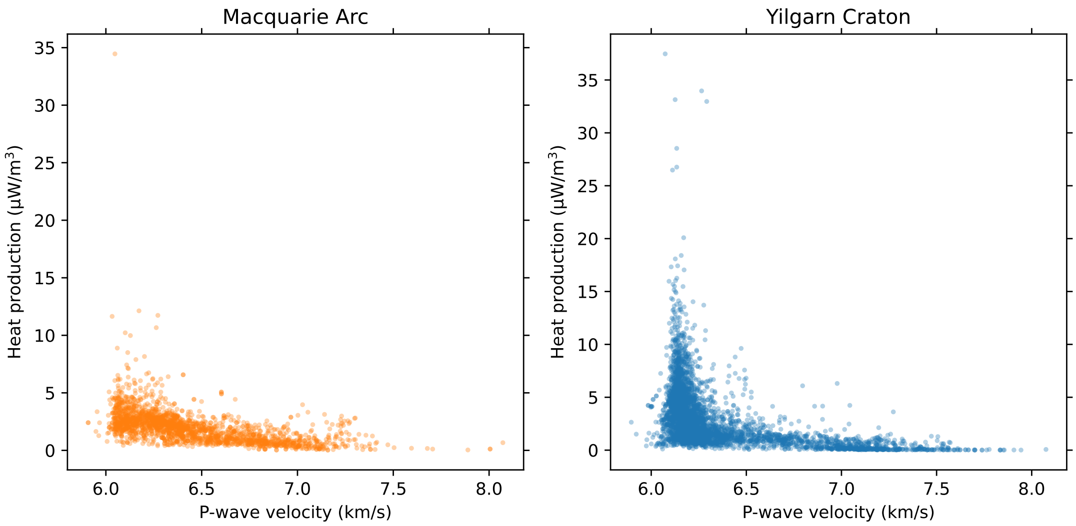


Figure 3: Heat production vs. V_p, linear space – Macquarie Arc and Yilgarn Craton.

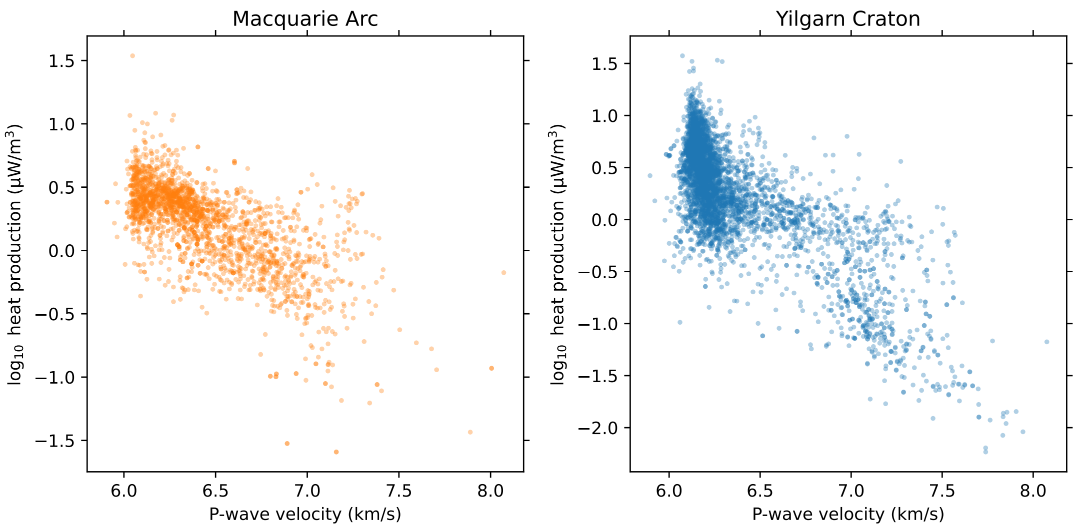


Figure 4: $\log_{10} A$ vs. V_p, same data, with the fitted terrane-specific relationships.

(c) Heat production is approximately log-normally distributed. Why would that fact alone explain why the same data looks convincing in one space and hopeless in the other?

Question 2. Universal Relationship?

The two terranes' fitted relationships are:

Macquarie Arc: $\log_{10} A = -0.727\,(V_p - 6.0) + 0.551$, Yilgarn Craton: $\log_{10} A = -1.193\,(V_p - 6.0) + 0.695$.

(a) What does it mean that the slopes are negative for heat production with composition?

(b) Which terrane's heat production is more sensitive to velocity, i.e., has the steeper slope?

(c) The global fit across all measured terranes gives a slope of −0.888, between these two. Is "one global relationship" a good description of what's actually going on, or is it closer to an average of many genuinely different local relationships?

Question 3. Does Sample Size Matter?

Fountain's original critique was based on a global compilation. The figure below shows r^2 against sample size N for 322 well-constrained terranes worldwide.

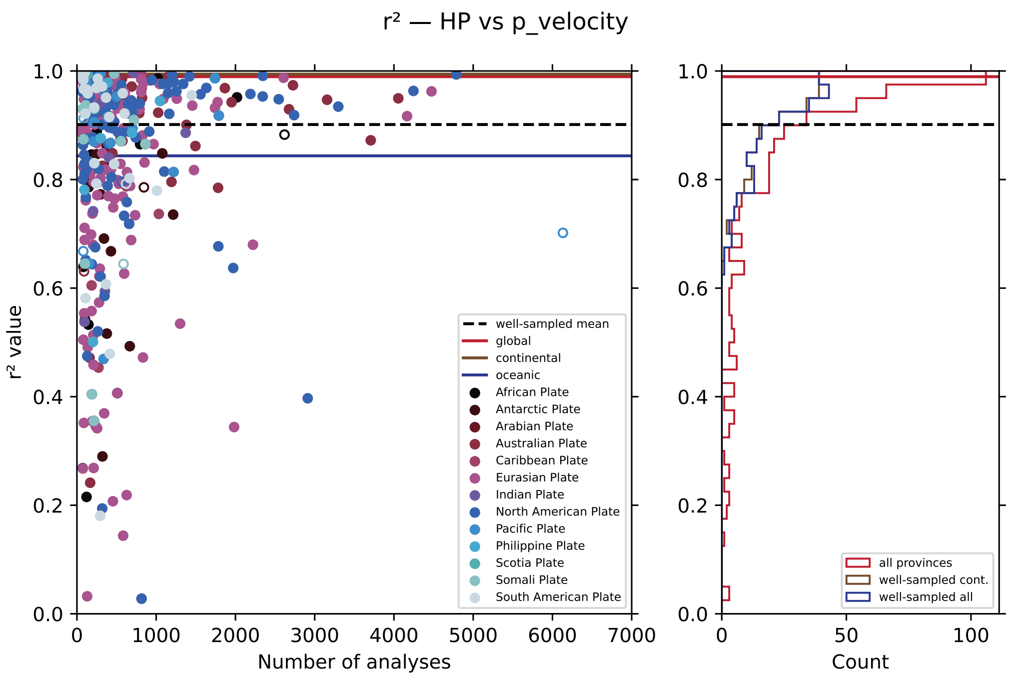


Figure 5: Goodness of fit (r^2) vs. sample size, across 322 terranes.

(a) Describe the shape of this plot. Does the scatter in r^2 stay constant as N grows, or change?

(b) If Fountain's compilation included many terranes with only a handful of samples each, how might that alone have made the relationship look weaker than it really is, without any of the log-space issue from Question 1?

Question 4. From Velocity to Heat Production

Table **??** gives median V_p in 5 km depth bins for both terranes, from a real crustal velocity model (AuSREM). For each bin, you will convert V_p into a heat production estimate and accumulate it into a crustal radiogenic heat flow.

Working in $\log_{10}$ space is faster: compute

$$\log_{10} A = m_0(V_p - 6.0) + m_1$$

first, then

$$A = 10^{\log_{10} A},$$

then multiply by the bin's thickness Δz (in km) to get that bin's contribution to heat flow, in $\mathrm{mW\,m^{-2}}$ — the units work out directly, with no conversion factor needed, because $1\ \mu\mathrm{W\,m^{-3}} \times 1\ \mathrm{km} = 1\ \mathrm{mW\,m^{-2}}$.

Worked example (first row, Macquarie Arc): at $V_p = 6.02$,

$$\log_{10} A = -0.727(6.02 - 6.0) + 0.551 = 0.537,$$

so

$$A = 10^{0.537} = 3.44\ \mu\mathrm{W\,m^{-3}}.$$

Over the 5 km bin, this contributes $3.44 \times 5 = 17.2\ \mathrm{mW\,m^{-2}}$. The Yilgarn Craton row is done the same way, using its own m_0 and m_1.

(a) Complete the table: fill in $\log_{10} A$, A, and each bin's contribution, then sum each terrane's column to get $q_{\mathrm{rad,median}}$.

Table 1: P-wave velocity statistics by depth interval for each terrane.

Depth (km)	N	V_p Median (km s^{-1})	V_p Mean (km s^{-1})	V_p Std (km s^{-1})	$\log_{10} A$	A
Macquarie Arc						
0–5	87	6.02	5.91	0.28	$-0.727\,(6.02-6.0)+0.551=0.537$	$10^{0.537}=3.44$
5–10	87	6.18	6.16	0.13		
10–15	87	6.31	6.32	0.13		
15–20	87	6.47	6.48	0.11		
20–25	87	6.62	6.63	0.09		
25–30	87	6.87	6.87	0.10		
30–35	86	7.10	7.10	0.13		
35–39.2	81	7.49	7.47	0.29		
40–42.6	48	7.65	7.63	0.25		
						$q_{\text{rad}}=$
Yilgarn Craton						
0–5	256	6.04	6.03	0.20	$-1.193\,(6.04-6.0)+0.695=0.647$	$10^{0.647}=4.436$
5–10	256	6.13	6.16	0.15	0.540	3.47
10–15	256	6.33	6.32	0.16	0.301	2.00
15–20	253	6.54	6.53	0.13	0.051	1.24
20–25	253	6.67	6.72	0.18	-0.104	0.787
25–30	253	6.94	6.98	0.20	-0.426	0.375
30–34.7	252	7.32	7.36	0.29	-0.880	0.132
35–38.3	184	7.79	7.81	0.20	-1.44	0.036
40–41.8	73	7.88	7.90	0.16	-1.55	0.028
						$q_{\text{rad}}=$

(b) Which terrane's crustal column contributes more heat flow by this estimate? Does that match what you'd have predicted from the slopes in Question 2 alone, or does the shape of the velocity profile with depth also matter?

Question 5. The Log-Normal Correction

The fitted relationship predicts the *median* A at a given V_p – but *heat production* is an additive, volumetric quantity, so the physically relevant value for the integral you just computed is the *mean*, not the median. Because A is log-normally distributed, the mean is always larger than the median, by a factor

$$f \approx 10^{1.151\sigma_{10}^2},$$

where σ_{10} is the scatter of $\log_{10} A$ about the fitted line. Measured directly from each terrane's own data: $\sigma_{10} = 0.247$ ($f = 1.175$) for Macquarie Arc, and $\sigma_{10} = 0.342$ ($f = 1.364$) for Yilgarn Craton.

(a) Multiply your two sums from Question 4 by the appropriate f to get the corrected, mean-based radiogenic heat flow estimate for each terrane.

Yilgarn Craton: ☐ $\mathrm{mW\,m^{-2}}$ Macquarie Arc: ☐ $\mathrm{mW\,m^{-2}}$

(b) Which terrane's estimate changed more in absolute terms? Why, given the two f values?

Question 6. A Paradox?

Real heat flow measurements exist for both terranes.

	Sites	Mean observed q ($\mathrm{mW\,m^{-2}}$)	Median observed q ($\mathrm{mW\,m^{-2}}$)
Macquarie Arc	48	71.2	70.3
Yilgarn Craton	66	37.8	37.1

In steady state, observed heat flow is the sum of everything generated in the crust above the measurement plus whatever is still arriving from the mantle: $q_{\text{observed}} = q_{\text{rad}} + q_{\text{mantle}}$, so $q_{\text{mantle}} = q_{\text{observed}} - q_{\text{rad}}$.

(a) For each terrane, using your corrected (mean-based) q_{rad} from Question 5, compute the implied mantle heat flow q_{mantle}.

(b) Is either result physically possible? What would a negative q_{mantle} actually mean?

(c) Two independent, individually reasonable methods, a velocity-based proxy calibrated against real rock samples and a direct energy-conservation argument, disagree here, sometimes severely. Which do you trust more, and why? What does each method actually depend on that the other doesn't?

(d) This activity opened by asking whether heat production could be estimated the way density, susceptibility, or resistivity can be. Having worked through this, what's your answer now. . . and what, specifically, makes heat production harder?

Key Ideas

A relationship can be real and still look like noise, if you're looking in the wrong space. Fountain's critique wasn't wrong about what he saw – linear-space A-V_p data genuinely is scattered. It was wrong about what that scatter meant, because it didn't account for heat production's log-normal distribution.

An empirical proxy relationship is only as trustworthy as the data used to build it though the relation can be effective without being universal. Different terranes show different, real, well-constrained slopes; treating any single fit (even the well-supported global one) as a universal law discards information the data itself is offering.

When two independently-motivated methods disagree, ask what each one is actually constrained by. A velocity-based proxy is constrained only by the rock samples used to calibrate it; nothing stops it from predicting an impossible total. An energy-conservation argument is constrained by the whole system having to balance, and cannot produce a physically impossible answer without signalling that fact directly, as a negative flux. That asymmetry, not just "geophysics being more careful," is why the energy-balance constraint deserves more trust when the two disagree.

Activity 8.E
Building a Geotherm

Purpose

Every steady-state geotherm you have interpreted so far in this workshop – the refraction problem, the layered/anisotropic conductivity problem, the velocity-based heat production estimate – has depended on a geotherm existing without ever building one from scratch. This activity does that: given $k(z)$, $A(z)$, a surface temperature T_0, and a surface heat flow q_0, you will construct a full steady-state geotherm by hand, in two separate stages, and follow it all the way to the mantle adiabat.

Learning Outcomes

By the end of this activity, you should be able to:

- List the ingredients needed to compute a steady-state geotherm, and identify which equation needs which of them.
- Compute $q(z)$ and $T(z)$ using a two-stage piecewise method, and explain why $q(z)$ can be completed before $T(z)$ is ever touched.
- Interpret the depth at which a conductive geotherm meets the mantle adiabat as the base of the thermal lithosphere.
- Predict and explain how changes in k and in A reshape a geotherm differently – including a case where *more* heat production produces *lower* temperatures.

Geological Context

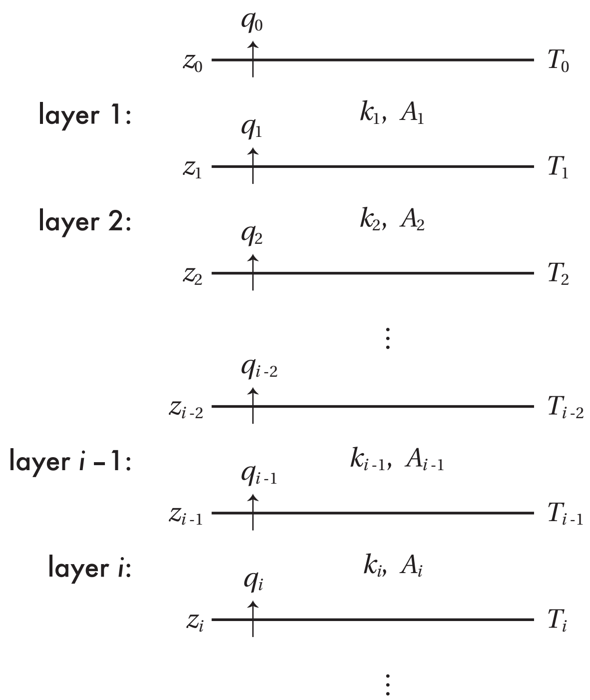


Figure 6: Layered Earth model for 1-D geotherms assuming constant physical property layers (or sublayers).

Assuming a layered Earth within which the properties thermal conductivity and heat production are constant (Figure **??**), we can write the above expressions as

$$q_i = q_i - 1 - A_i(z_i - z_{i-1}) \quad \text{where } x \in [1 \dots N],$$

and

$$T_i = T_{i-1} + \frac{q_{i-1}}{k_i}(z_i - z_{i-1}) - \frac{A_i}{2k_i}(z_i - z_{i-1})^2,$$

where i denotes the layer and N the total number of layers. We can further simplify the calculations by defining $\Delta z_i = z_i - z_{i-1}$. Note that temperature and heat flow are determined at the top and bottom of and the physical properties are defined within each layer.

Notice that the first equation involves only $A(z)$ and q_0 – not k or T. The second equation needs $q(z)$, which means it cannot be started until the first is finished. That ordering is not a computational convenience; it is a real physical fact about which quantities depend on which.

Eventually, a conductive geotherm should meet the **mantle adiabat**, which can be approximated as $T = 1300 + 0.3z$ (z in km, T in °C). The adiabatic temperature profile is the temperature path mantle material follows during convection, which controls how heat moves. The depth these two temperature paths cross marks the base of the thermal lithosphere.

Tasks

Question 1. The Recipe

(a) From the two equations above, list everything you need to know before you can compute a geotherm.

(b) Which of those quantities does $q(z)$ actually depend on? Which does $T(z)$ depend on that $q(z)$ does not?

Question 2. Stage 1: Building $q(z)$

Our crustal column has the layers below, with surface boundary conditions: $q_0 = 60\ \mathrm{mW\,m^{-2}}$ and $T_0 = 15\ \circ$C.

Layer	Depth range km	k_i $\mathrm{W\,m^{-1}\,K^{-1}}$	A_i $\mu\mathrm{W\,m^{-3}}$
upper crust	0–15	3.0	1.5
lower crust	15–40	2.1	0.2
upper mantle	40+	3.5	0.02

Complete the table below. At each step, using equation **??** and A_i of whichever layer that step falls in. The first step is worked for you.

z_i km	Layer	Δz_i km	$A \cdot \Delta z_i$ $\mathrm{mW\,m^{-2}}$	q_i $\mathrm{mW\,m^{-2}}$
0	–	–	–	60.0
5	upper crust	5	7.5	52.5
10	upper crust	5		
15	upper crust	5		
40	lower crust	25		
80	upper mantle	20		
100	upper mantle	20		

Question 3. Stage 2: Building $T(z)$
Now use your completed $q(z)$ column to estimate temperatures using Equation **??**. The first step is worked for you.

z_i km	Δz_i km	k_i $\mathrm{W\,m^{-1}\,K^{-1}}$	A_i K	q_i $\mathrm{mW\,m^{-2}}$	T_i °C
0	–	–	–	60.0	15.0
5	5	3.0	1.5	52.5	108.75
10	5	3.0	1.5		
15	5	3.0	1.5		
40	25	2.1	0.2		
80	20	2.1	0.02		
100	20	2.1	0.02		

Your temperature result at 40 km depth is the Moho temperature.

Question 4. Plotting the Geotherm
Plot your Stage 1 & 2 results along with the table below which fills in some of the gaps. Draw a curve through the dots. Also plot the mantle adiabat $T = 1300 + 0.3z$ (two points are enough since it's a straight line). A more realistic adiabat is slightly more complex, but not by much.

z_i km	Layer	$q(z_i)$ $\mathrm{mW\,m^{-2}}$	$T(z_i)$ °C
20	lower crust	36.5	347
25	lower crust	35.5	433
30	lower crust	34.5	516
35	lower crust	33.5	597
50	mantle lithosphere	32.3	768
60	mantle lithosphere	32.1	860
70	mantle lithosphere	31.9	951
90	mantle lithosphere	31.5	1133
100	mantle lithosphere	31.3	1222
110	mantle lithosphere	31.1	1311

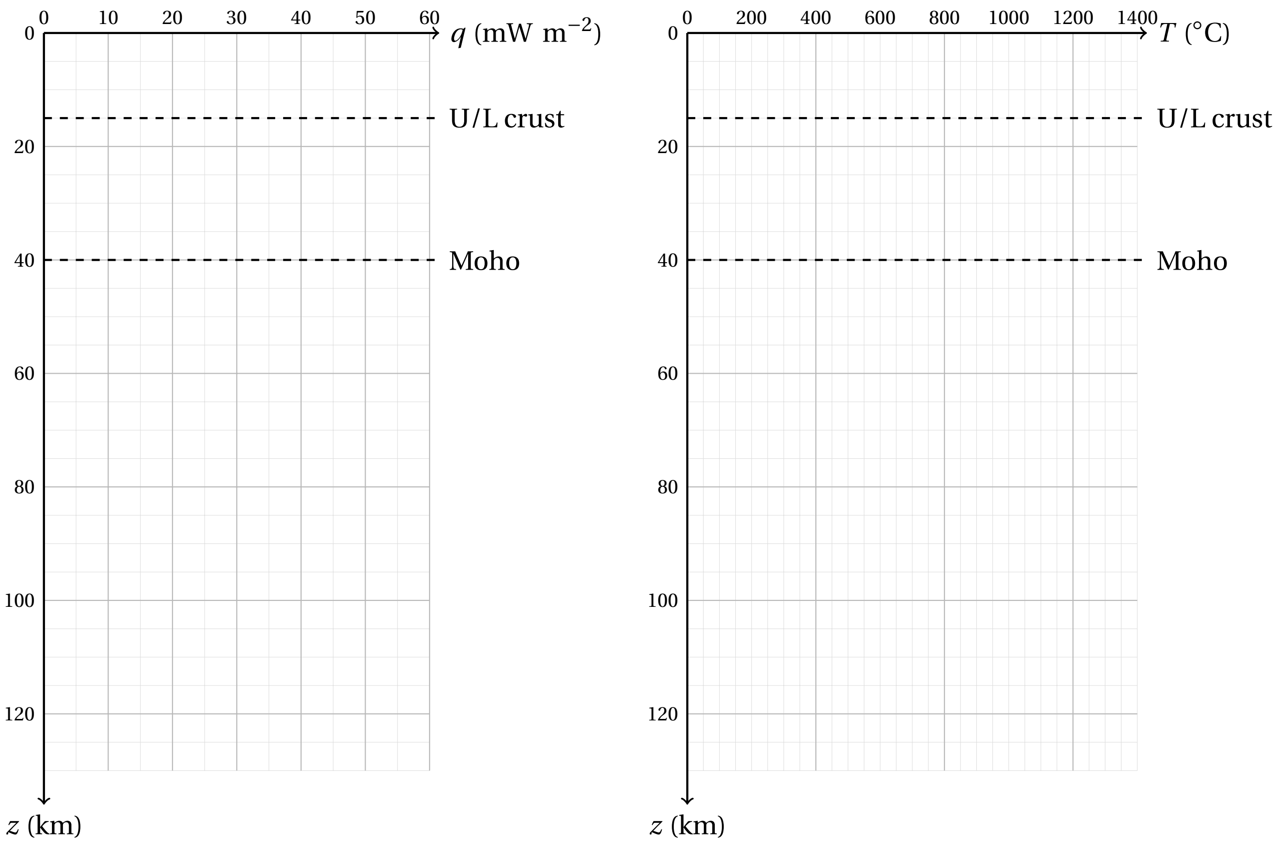


Figure 7: Axes for plotting heat flow and geotherms.

(a) How does heat flow change between the crustal layers when heat production is constant within the layer?

(b) Where is the temperature curvature most visible on your plot – and where is the curve closest to a straight line? Does that match which layers have the largest A?

(c) What does the depth where your geotherm meets the adiabat physically represent?

Question 5. Changing k

Recompute *only* $T(z)$ (not $q(z)$ as $A(z)$ has not changed) using $k_1' = 1.5\ \mathrm{W\,m^{-1}\,K^{-1}}$ for the upper crust instead of 3.0. Plot the result on the same axes above.

z_i km	Δz_i km	q_i $\mathrm{W\,m^{-1}\,K^{-1}}$	k_i K	A_i $\mathrm{mW\,m^{-2}}$	T_i °C
0	–	–	–	60.0	15.0
5	5	1.5	1.5		
10	5	1.5	1.5		
15	5	1.5	1.5		
40	25	2.1	0.2		
80	20	2.1	0.02		
100	20	2.1	0.02		

How does a decrease in thermal conductivity affect the geotherm? What would happen if you were to increase it instead?

Question 6. Changing A

Now recompute *both* $q(z)$ and $T(z)$ using $A_1' = 3.0\ \mu\mathrm{W\,m^{-3}}$ for the upper crust instead of 1.5 (keep $k_1 = 3.0$). Plot the result on the same axes above (label the models).

z_i	Δz_i	q_i	k_i	A_i	T_i	
km	km	W m^{-2} K^{-1}	km	K	mW m^{-3}	°C
0	–	–	–	60.0	15.0	
5	5	3.0	3.0			
10	5	3.0	3.0			
15	5	3.0	3.0			
40	25	2.1	0.2			
80	20	2.1	0.02			
100	20	2.1	0.02			

How does a decrease in thermal conductivity affect the geotherm? What would happen if you were to increase it instead?

Question 7. Contemplate, Then Check

(a) If the k change in Question 5 persisted through the whole column, would you expect the Moho temperature and the lithosphere thickness (adiabat-crossing depth) to increase or decrease relative to the base case? Explain your reasoning before looking below.

(b) Now do the same for the A change in Question 6 – predict the direction *before* checking.

(c) **Reveal:** continuing each perturbation through the full column gives:

	Moho temperature (°C)	Adiabat crossing (km)
Base case	675	113
$k'_1 = 1.5$ (halved)	919	84
$A'_1 = 3.0$ (doubled)	351	never, within any plausible depth

Was your prediction for the A case right? If not, what does doubling A_1 actually do to $q(z)$

everywhere below 15 km, and why does that matter more than the extra heat generated in the upper 15 km itself? And now that you've seen a geotherm that never reaches the adiabat at all – what would that actually mean, physically, if it were real?

Key Ideas

$q(z)$ **and** $T(z)$ **depend on different things, in a strict order.** $q(z)$ comes from $A(z)$ and q_0 alone; $T(z)$ needs $q(z)$ as an input. That one-way dependency is why the recipe works layer by layer instead of all at once.

Where a geotherm crosses the mantle adiabat marks the base of the thermal lithosphere. Below that depth, material convects and follows the adiabat; above it, conduction sets the temperature.

Whether more heat production makes a column hotter or colder depends on which boundary condition is fixed. Hold surface heat flow q_0 fixed, and near-surface heat production can *lower* temperatures at depth; hold the deep heat flow fixed instead, and the same change raises them. The direction of the effect isn't a fixed property of A – it follows from what you're allowed to hold constant.

A geotherm that never reaches the convecting adiabat is telling you something is wrong with the inputs. Generally such a result points to too high a heat production or too low a heat flow. Likewise, if we compute temperatures that would suggest widespread melting in the mid to lower crust, unless we are in a magmatic arc, it is probably indicates our surface heat flow is too high or heat production too low. *The model breaking is the useful result.*